

The convinced atheist objection

The hypothesis that an intelligent designer exists is really, really unlikely in the first place, so premise (3) is false.

the fine-tuning argument

1. $Pr(e \mid \text{design}) = 0.5$
2. $Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$
3. $P(\text{design}) = P(\text{non-design}) = 0.5$.
4. Bayes' theorem.
5. $P(\text{design}) \approx 1$ (1,2,3)
6. If the universe was created by an intelligent designer, then God exists.

C. $P(\text{God exists}) \approx 1$.

Skepticism about physics

The physics is wrong (or at least might be).

The anthropic objection

We could never observe the falsity of the claim that the constants permits life since, if it were false, we would not exist to observe it.

Let's look at one last objection:

The multiverse objection

There are very many — perhaps infinitely many — distinct universes, which can have different fundamental physical constants.

Surely, this objection goes, if there were enough universes, then no matter how improbable it is that one of them would have constants that fall in the life-permitting range, it is not so improbable that some would.

Our situation would then be somewhat analogous to the position of someone who wins the lottery. The odds of **that** person winning the lottery were very small; but if enough people buy tickets, it is not so improbable that someone wins.

So the key question is: do we have good reason to think that the multiverse hypothesis is true?

A first point to note: it would be very surprising if this hypothesis were true. For, if it is, there are very many — perhaps infinitely many — other universes, each as real as ours, in which some near-duplicate of you exists. There is, for example, very likely one in which there exists some being with a qualitatively identical history to you who differs from you only in that she or he scratched their nose one second ago.

This does not show that the multiverse hypothesis is false; the universe might be strange, and science has often shown us that it is. But it does suggest that the multiverse hypothesis is not one that we should believe without argument.



fine-
tuning of the
universe

A solid red circle with a subtle drop shadow, centered on a white background. Inside the circle, the text "Bayes' theorem" is written in a white, sans-serif font, centered horizontally and vertically.

Bayes'
theorem



the
argument



objections

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Enough about probability (for now). Let's turn now to some results from

follows.

contemporary physics which will be important to the argument which

A first question: what does the theory provide by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)

These 'dimensions' will be our focus.

One such constant is the cosmological constant, which measures the

energy density of empty space.

could have different values consistent with the laws of physics.

The cosmological constant is just a number which (as far as we know)

(like the cosmological constant) would fall in a certain range.

One thing that the standard model of physics gives us is a measure of how

likely it is, given the laws of the nature, that the fundamental constants

life to have evolved. For example, if there were nothing but hydrogen, it is

hard to see how life could have evolved. If there were no planets, it is hard

to see how life could have evolved.

We can make certain plausible assumptions about what it would take for

constant has a value which would permit the evolution of life.

Given assumptions such as these, we can look at what the standard model

of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement.

(There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

supporting value, given the laws of nature, is very low.

This is the claim that the cosmological constant having a life-

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

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The claim is that, according to current physics, the probability of the



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1



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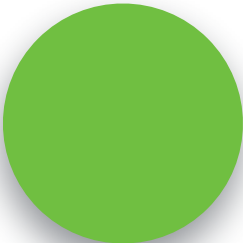


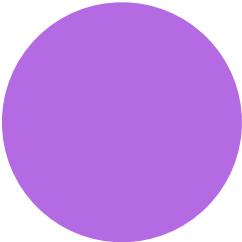
fine-
tuning of the
universe

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Bayes'
theorem





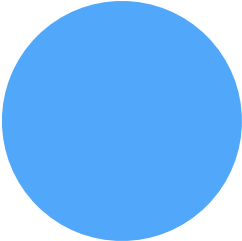








the
argument



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Let's now ask how to turn these facts about current physics and

probability theory into an argument for the existence of God.

in the turn, so we can consider two different hypotheses about the origins

Much as we considered two different hypotheses about the number of balls

of the universe.

created by an intelligent designer. Let the non-design hypothesis be the

Let the design hypothesis be the hypothesis that the universe

hypothesis that it was not.

value in the life-permitting range.

Let e be the fact that the cosmological constant has a

Then what do we know about relevant probabilities?



$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$Pr(e | \text{design}) \equiv 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

With this information in hand, figuring out the probability of the non-

numbers.

design hypothesis given evidence is that just plugging in the



$$Pr(\text{non-design}) = 0.5$$



$$Pr(\text{design}) = 0.5$$



$$Pr(e) = 0.25$$



$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

PowerBar®

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now



consider the odds of winning Powerball one trillion times in a row. Call that a “super

It is difficult to think about numbers as large as the denominator of this fraction. But

Now consider the odds of winning a super Powerball one trillion times in a row. Call

that a 'super duper Powerball.'

Now consider the odds of winning a super Powerball one trillion times in a row.

odd of the universe being life-permitting by chance.

The odds of this happening are about $1/10^{44}$ — so much, much higher than the

This means that if you simply take the physics at face value, and begin by assigning a

the non-design hypothesis being true are vastly lower than the chances of winning a

superduperpowerball a trillion times in a row.

probability of 0.5 to the non-design hypothesis, you should think that the chances of

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1.

minus the very small number we're just discussing:



$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This about as close to certainty as it is possible to get.



objections



6. *Life exists.*

$$C.P(God\ exists)\approx 1.(7,8)$$



2



P

$$7. P(\text{design}) \approx 1. \quad (5, 6)$$

4. Bayes', theorem.

5



P

• (1, 2, 3, 4)



8. If the universe was created by an

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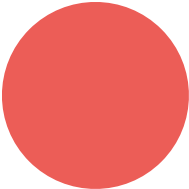
intelligent designer, then God exists.

$$(life / design) \equiv 0.5$$

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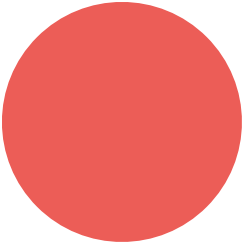
$$(\text{design} / \text{life}) \approx 1$$







fine-
tuning of the
universe



should not take the fact that we are in a life-permitting zone to be evidence for much

of anything.

Just so, if we are confident that there are a huge number of different universes, we

is, there are very many — perhaps infinitely many — other universes, each as real as

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